Cohort Reconstruction Methodology

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Introduction

Efforts to implement age-structured cohort reconstruction (CR) of Sacramento River fall-run Chinook Salmon (SFCS) populations are under way. The currently used Sacramento Index (SI) serves as an age-aggregated index of adult ocean abundance and potential adult escapement [1]. The CR efforts aims to update this method by reporting age-specific assessments of hatchery and natural-origin escapement with estimates of uncertainty. Accurate estimates of age-specific escapement for can be used to estimate maturation rates, mortality rates, and rates of exploitation. The hatchery-origin component of escapement is estimated using recovered coded-wire-tag (CWT), but the natural-origin component is reconstructed by aging scales collected from untagged fish. At the end of each spawning season survey crews will be expected to contribute relevant data and a sample of scales from untagged fish that will be pooled with other locations to produce basin-wide estimates of natural-origin age classes.

This document aims to provide guidance to Sacramento River Basin (SRB) escapement surveys on collecting scales for aging and providing the data needed to produce estimates of abundance for each age-class of natural and hatchery-origin SFCS in the SRB. It also provides the code and steps necessary to take the survey and scale age data and produce the final estimates of cohort abundance.

Ensuring Unbiased Scale Recovery

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Data needs

At the end of escapement surveys in basin natural spawning grounds and hatcheries, each survey group should be able to provide the following data to the basin coordinator:

1. $\hat{N}_l$ - The estimate of abundance for each location l in the basin. For hatcheries this value should be close to exact, but for carcass and other spawning ground surveys there should be uncertainty in this estimate. Currently the uncertainty of $\hat{N}_l$ from carcass surveys is not reported, but it can be calculated using the current escapement estimate methods and we should consider incorporating it into cohort reconstruction efforts.

2. n_l - The number of fish recovered and checked for tags (whether CWT or PBT) for location l.

3. $\hat{\theta}_l$ - Theta, the sampling fraction of the survey, or the probability of recovering a tag. Calculated as $\frac{n_l}{\hat{N}_l}$. For hatcheries $\hat{\theta}_l$ should be 1.

4. F_{prod} - A table that includes records for each recovered release groups with the following fields:

   - `release_group` : a unique identifier for each recovered release group designated by tagging.
   - `location` : the location of the survey.
   - `tag_rate` : the tagging rate of the release group.
   - `return_year` : the survey or escapement year.
   - `brood_year` : the year the tagged fish was released.
   - `tag_age` : age of release group, calculated as `return_year` - `brood_year`.
   - `brood_location` : location that the release group originates from.
   - `frequency` : the number of fish from this release group recovered in the survey.

Table 1: Example of F_prod table for simulated recoveries on Clear Creek.

location	release_group	tag_rate	return_year	brood_year	tag_age	brood_location	frequency
Clear	American 2017	0.25	2020	2017	3	American	4
Clear	Clear 2016	0.25	2020	2016	4	Clear	43

location	release_group	tag_rate	return_year	brood_year	tag_age	brood_location	frequency
Clear	Clear 2017	0.25	2020	2017	3	Clear	62
Clear	Clear 2018	0.25	2020	2018	2	Clear	14
Clear	FRH 2016	0.25	2020	2016	4	FRH	1
Clear	FRH 2017	0.25	2020	2017	3	FRH	1

Estimating the abundance of hatchery-origin fish

Escapement to hatcheries and spawning grounds is composed of both tagged and untagged hatchery-origin fish as well as untagged natural-origin fish. Once the required data from each spawning survey location has been collected by the basin coordinator, the tag recovery data can be used to estimate the abundance of each age-class for tagged and untagged hatchery-origin fish in that locations spawning population.

For each recovered tag an estimate of the number of unrecovered tags, $\hat{k}$, can be made by drawing from a negative binomial distribution where $\hat{\theta}_l$ is the probability of recovering that tag:

$$\hat{k} \sim NB(1, \hat{\theta}_l)$$

This process is iterated 1000 times for each tag release group recovered to produce estimates of uncertainty in our final cohort estimates. For the `F_prod` table provided by each location this can be done with the `tag_k_draws()` function:

```
#select clear creek data from loc_data
fprod<-loc_data[["Clear"]]$F_prod
theta<-loc_data[["Clear"]]$theta
N_loc<-loc_data[["Clear"]]$N_loc
clr_creek_list<-unique(fprod$release_group) #pull list of release_groups
clr_creek_results<-data.frame() #prepare dataframe to store results

#run for-loop over each release group
for(uid in clr_creek_list){
  tag_data<-fprod%>%
    filter(release_group==uid)
  #run tag_k_draws() for each release_group
  tag_outputs<-tag_k_draws(tag_data=tag_data,
                           theta=theta,
                           iterations=1000,
```

```
                              freq=tag_data$frequency,
                              f_prod=tag_data$tag_rate)
  clr_creek_results<-clr_creek_results%>%rbind(tag_outputs) #store results
}
```

Above is an example of running the iterative k-draws for each of ten simulated release groups recovered on Clear Creek in 2020. The below output `clr_creek_results` contains 10,000 records (10 release groups x 1000 iterations) for each iteration and includes the estimate of:

- `total_tags` - $\hat{N}_{h,tagged}$, the k-draw iterations' estimate of the number of tagged hatchery-origin fish in the spawning population, including those recovered and unrecovered.
- `total_hatchery` - $\hat{N}_h$, the k-draw iterations' estimate of the abundance of hatchery-origin fish in the spawning population. Estimated by expanding `total_tags` by `tag_rate`.

Table 2: Example of k-draw output for simulated recoveries on Clear Creek.

release_group	location	theta	tag_rate	tag_freq	total_tags	total_hatchery	k_iteration
American 2017	Clear	0.13	0.25	4	39	156	1
American 2017	Clear	0.13	0.25	4	39	156	2
American 2017	Clear	0.13	0.25	4	16	64	3
American 2017	Clear	0.13	0.25	4	81	324	4
American 2017	Clear	0.13	0.25	4	52	208	5
American 2017	Clear	0.13	0.25	4	36	144	6

The above k-draw outputs can be performed for each survey location and saved for later use to allow us to characterize uncertainty in our final estimates. First though we need to determine how many scales each spawning survey location should contribute to produce our final pooled basin-wide estimate of age-class abundance.

For each k-draw iteration for the above Clear Creek data we can estimate the total number of recovered tags (n_{tagged}), the total number of tagged fish in the spawning population ($\hat{N}_{tagged}$), and the total number of hatchery origin fish in the spawning population ($\hat{N}_{hatchery}$).

```
clr_creek_estimates<-clr_creek_results%>%
  group_by(k_iteration,location,return_year,theta)%>%
    summarise(recovered_tags=sum(tag_freq),
              total_tags=sum(total_tags),
              total_hatchery=sum(total_hatchery))
```

With the above we can calculate the average of $\hat{N}_{tagged}$ (`total_tags`), $\hat{N}_{hatchery}$ (`total_hatchery`), and $\hat{N}_{untagged}$ the number of untagged fish (`N_untagged`) for the simulated Clear Creek spawning population:

```
clr_creek_estimates<-clr_creek_estimates%>%
  group_by(location,return_year,
           recovered_tags)%>%
  summarise(total_tags=round(mean(total_tags)),
            total_hatchery=round(mean(total_hatchery)))%>%
  mutate(N_loc=N_loc)%>% #add location total escapement
    mutate(N_untagged=N_loc-total_tags) #estimate N untagged
```

Table 3: Example of hatchery-origin abundance estimates for simulated recoveries on Clear Creek.

location	return_year	recovered_tags	total_tags	total_hatchery	N_loc	N_untagged
Clear	2020	135	1036	4090	6631	5595

Once the above calculations have been made for each surveyed spawning population the basin coordinator has a table that shows the above for each location:

Table 4: Example of hatchery-origin abundance estimates for simulated recoveries at several survey locations.

location	return_year	recovered_tags	total_tags	total_hatchery	N_loc	N_untagged
American	2020	2133	3743	14839	36327	32584
Clear	2020	135	1035	4087	6631	5596
Feather	2020	868	7903	29644	42724	34821
Sac	2020	190	1461	5832	13526	12065
Yuba	2020	10	164	569	4194	4030
FRH	2020	5309	5309	21208	25718	20409

Obtaining scales for aging from each survey

The purpose of aging scales of untagged fish is to estimate the abundance of each age class of untagged fish, and then for natural-origin fish. To this each survey location will be expected to submit a sample of scales collected from untagged fish for aging to contribute to a pool of all SRB fish. The estimation of $\hat{N}_{l,untagged}$ produced above is necessary as it allows us to accurately estimate how many scale samples we will need to age from each location for an unbiased estimate of age-class abundance for natural-origin SFCS. Different spawning locations can have very different compositions of hatchery and natural-origin fish, as well as different proportions of each age-class. If the pool of scales for the SRB is not representative of all the locations, over or under-representation in some locations can cause major bias in our final estimates.

The basin coordinator can estimate the proportion of untagged fish ($p_{untagged}$) in the SRB that are contributed by each location by dividing $\hat{N}_{l,untagged}$ by $\sum \hat{N}_{l,untagged}$:

```
#estimate N_untagged proportion from each survey
basin_untagged<-sum(loc_results$N_untagged)
loc_results<-loc_results%>%
  mutate(untagged_prop=N_untagged/basin_untagged)
```

Table 5: Estimates of each locations proportional contribution to the basin-wide untagged SFCS population.

return_year	location	N_untagged	untagged_prop
2020	American	32584	0.270793165
2020	Clear	5596	0.046506216
2020	Feather	34821	0.289384017
2020	Sac	12065	0.100267602
2020	Yuba	4030	0.033491789
2020	FRH	20409	0.169611396
2020	NFH	5356	0.044511668
2020	Battle	4488	0.037298052
2020	Other	979	0.008136095

With an estimate of $\hat{p}_{l,untagged}$ the number of scales each locations needs to contribute s_l can be calculated as:

$$s_l = S_b * \hat{p}_{l,untagged}$$

Where S_b is the number of scales to be aged for the entire SRB. For our example $S_b = 500$, and we can use the `round_to_sum()` function, which was created to ensure that any rounding errors from this calculation are accounted for and allocates remainders.

```
S_b=500 #set basin scale sample size
loc_results$target_scales=round_to_sum(loc_results$untagged_prop,S_b)

#create a list of fish recovered and fish recovered w/out tags from each location
loc_recovered<-data.frame(
  location = names(loc_data),
  n_recovered = sapply(loc_data, function(x) x$n_recovered),
  n_nocwt=sapply(loc_data,function(x) x$n_nocwt)
)

#join to loc_results
loc_results<-loc_results%>%
  left_join(loc_recovered)
```

The below table shows $\hat{N}_{l,untagged}$ (`N_untagged`, estimate of abundance of untagged fish in that locations survey), $n_{l,untagged}$ (`n_nocwt`, number of fish recovered without tags), $\hat{p}_{l,untagged}$ (`untagged_prop`), and s_l (`target_scales`) for each survey location.

Table 6: Shows the number of scales each location is expected to contribute when the basin-wide scale sample size is 500.

return_year	location	N_untagged	n_nocwt	untagged_prop	target_scales
2020	American	32584	18573	0.270793165	135
2020	Clear	5596	727	0.046506216	23
2020	Feather	34821	3832	0.289384017	145
2020	Sac	12065	1568	0.100267602	50
2020	Yuba	4030	242	0.033491789	17
2020	FRH	20409	20409	0.169611396	85
2020	NFH	5356	5356	0.044511668	22
2020	Battle	4488	898	0.037298052	19
2020	Other	979	196	0.008136095	4

It's important here that the basin coordinator ensure that for no location s_l is greater than $n_{l,untagged}$. This should only occur when S_b is sufficiently large, typically greater than 5000, however if for any location this happens, it can result in that location being under-represented in the aged scale pool and resulting in biased cohort estimates. If this does occur the fix is to adjust S_b to the limiting location. In the below example of this, we can find the maximum value of S_b for this simulated population.

```
loc_results$max_based_on_loc<-loc_results$n_nocwt/loc_results$untagged_prop
limiting_location <- loc_results$location[which.min(loc_results$max_based_on_loc)]

adjusted_S_b<-floor(min(loc_results$max_based_on_loc))

print(paste("limiting location:",limiting_location))
```

```
[1] "limiting location: Yuba"
```

```
print(paste("maximum basin scale sample size:",adjusted_S_b))
```

```
[1] "maximum basin scale sample size: 7225"
```

Once s_l is calculated for each survey location, survey staff should select a subsample of scales for aging from those collected during the survey season. Scales for aging should be randomly selected from those available to ensure a random and representative subsample. For this walkthrough we can use a dataset that containts simulated recoveries for each of the surveyed locations:

```
#load location recoveries
loc_recoveries<-readRDS("outputs/loc_recoveries.rds")
head(loc_recoveries$Clear)%>%
  select(location,tag_age,origin,has_cwt,release_group)%>%
  flextable()%>%
  set_table_properties(width=1,layout = "autofit")%>%
  theme_vanilla() %>%
  fontsize(size = 10) %>%
  set_caption("Example of fish recoveries from Clear Creek, each record is a simulated recove
```

Table 7: Example of fish recoveries from Clear Creek, each record is a simulated recovered fish.

location	tag_age	origin	has_cwt	release_group
Clear		natural	0	

location	tag_age	origin	has_cwt	release_group
Clear	3	hatchery	1	Sac 2017
Clear	2	hatchery	1	Clear 2018
Clear		natural	0	
Clear		hatchery	0	
Clear		hatchery	0	

Below we can simulate the random selection of scales from each of the survey locations. Once a survey has selected their scales, they should be sent into the basin coordinatorfor aging **(Arthur note: this will need to be determined and detail added about scale aging process)**. We can then pool all the aged scales together into one data frame `basin_scales`.

```
#build loc_scales list for filling
loc_list<-names(loc_data)
loc_scales<-vector(mode = "list", length = length(loc_list))
names(loc_scales)<-loc_list

#for each location
for(l in loc_list){
  d<-loc_recoveries[[l]]
  scales_n=loc_results%>% #get target scale sample size
    filter(location==l)
  d<-d%>%
    filter(has_cwt==0) #only getting scales from untagged fish
  d_sample<-sample_n(d,scales_n$target_scales,replace = F) #sample scales
  d_sample<-d_sample%>%mutate(scale_age=return_year-brood_year)
  loc_scales[[l]]=d_sample
}

#pool basin scales together
basin_scales<-data.frame()
for(l in loc_list){
  basin_scales<-basin_scales%>%rbind(loc_scales[[l]])
}
```

Table 8: Example of aged scales pooled across the basin.

location	return_year	scale_age
American	2020	4
American	2020	3
American	2020	3
American	2020	4
American	2020	4
American	2020	3

Reconstructing Cohort Abundances

Once the scales have been aged and results sent to the basin-coordinator the process of cohort reconstruction can begin. First we can utilize the k-draws results from all the locations produced earlier by the `tag_k_draws()` function and join them into a single data frame `loc_hatcheries`.

```
#first take loc_kdraws and get basin wide estimate of hatchery origin fish for each iteration
loc_hatcheries<-data.frame()
for(l in loc_list){
  d<-loc_kdraws[[l]]
  d<-d%>%
    group_by(return_year,tag_age,location,k_iteration,theta)%>%
    summarise(tag_freq=sum(tag_freq),
              total_tags=sum(total_tags),
              total_hatchery=sum(total_hatchery))
  loc_hatcheries<-loc_hatcheries%>%rbind(d)
}

#sometimes locations have no hatchery fish so remove here
loc_hatcheries<-loc_hatcheries%>%filter(!is.na(tag_age))
```

`loc_hatcheries` contains the estimates of total abundance of hatchery-origin fish in each age class for each survey location, across all 1000 k-draw iterations, which allows us to characterize the uncertainty in our tag expansions.

With the `loc_hatcheries` data we can then sum across locations for estimates of basin-wide hatchery-origin abundance of each age class for each k-draw iteration:

```
#sum location based hatchery estimates from k-draws into basin-wide
basin_hatchery_estimates<-loc_hatcheries%>%
  group_by(return_year,tag_age,k_iteration)%>%
  summarise(tag_freq = sum(tag_freq),
            total_tags = sum(total_tags),
            total_hatchery = sum(total_hatchery))
```

Table 9: k-draw iteration estimates of abundance of hatchery-origin fish in each age class for the entire Sacramento R. basin.

return_year	tag_age	k_iteration	total_hatchery
2020	2	1	4406.154
2020	3	1	61808.000
2020	4	1	15698.026
2020	2	2	4102.615
2020	3	2	60792.000
2020	4	2	16374.500

Finally we can utilize the function `basin_CR()` to get estimates of age-class abundance along with estimates of uncertainty. The function requires the following inputs:

- `basin_N` - $\hat{N}$ the estimate of total abundance for all fish in the basin.
- `spawning_results` - a list containing two objects: the `basin_scales` data frame of scales sampled and aged, and the `hatchery_ages` data frame created above.
- `iterations` - number of bootstrap iterations for uncertainty, needs to match iterations of k-draws, default is 1000.
- `CI`- set target width of confidence intervals, default 95%.

Once inputs are built they can be run through the `basin_CR()` function, which can take up to 10 seconds to run, depending on the number of scales aged and iterations run.

```
#set function inputs
basin_N<-sum(sapply(loc_data, function(x) x$N_loc)) #sum all locaion N_hat
spawning_results<-list("basin_scales"=basin_scales,
                       "hatchery_ages"=basin_hatchery_estimates)
iterations=1000
```

```
CI=0.9

cr_results<-basin_CR(basin_N=basin_N,
                     spawning_results=spawning_results,
                     iterations=iterations,
                     CI=CI)
```

The above `cr_results` shows the outputs the following results for each age class a :

- `mean_hatchery` - $\hat{N}_{h,a}$ the average estimate of hatchery-origin fish across iterations.
- `lower_CI_hatchery/upper_CI_hatchery` - lower and upper confidence limits for $\hat{N}_{h,a}$.
- `mean_natural` - $\hat{N}_{n,a}$ the average estimate of natural-origin fish across iterations.
- `lower_CI_natural/upper_CI_natural` - lower and upper confidence limits for $\hat{N}_{n,a}$.
- `n_iterations` - number of iterations ran to characterize uncertainty.

Table 10: Final abundance estimates, with uncertainty, for each age class of natural-origin SFCS.

age	mean_natural	lower_CI_natural	upper_CI_natural
2	7189.538	4697.254	9649.433
3	29502.713	24784.001	34174.557
4	24041.572	19967.187	28328.420

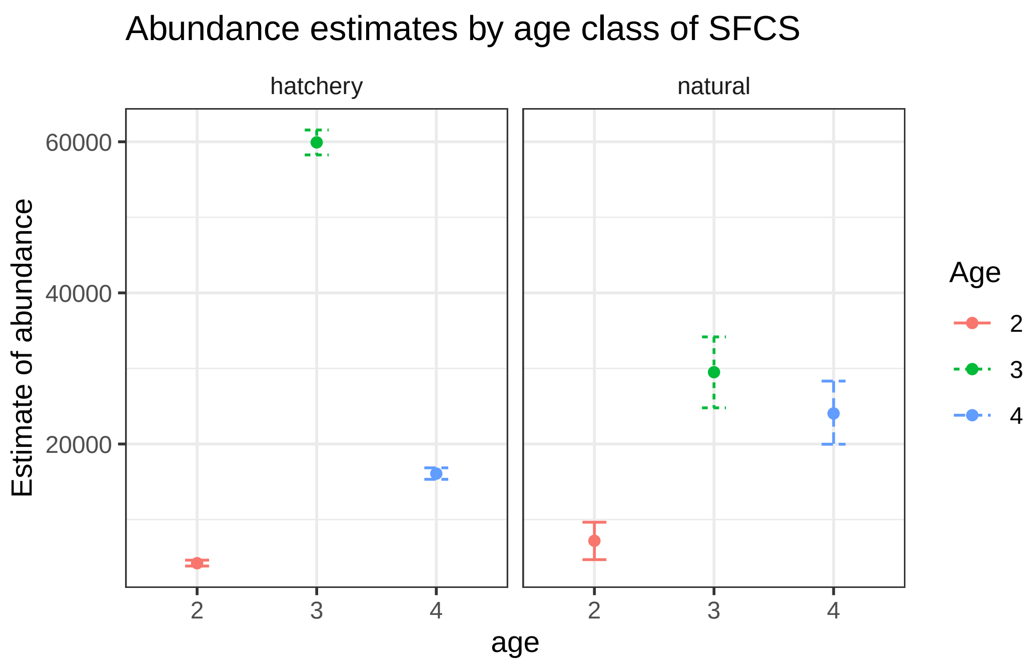

Figure 1: Plots showing estimates of age composition for natural and hatchery-origin SFCS, with upper and lower confidence intervals

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Appendix

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References

[1] Emily Chen et al. "Cohort Reconstruction for Sacramento River Fall Chinook salmon and Comparison with the Sacramento Index". In: *Pacific Fishery Management Council* (Sept. 2024).